

Computer Task X1 – Modelling Load Application

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1. Introduction

This report investigates three different methods for modelling load application on a cantilever beam-type structure using the Finite Element Method (FEM) in ANSYS Mechanical. The beam is loaded through a circular hole near its free end by a rigid cylinder that pushes downward, causing the beam to bend. The problem is treated as a static, linear-elastic, two-dimensional plane stress analysis.

The three methods compared are:

1. **Remote Force** – a downward force applied to a virtual point at the hole centre, with four internal distribution behaviours: Coupled, Rigid, Deformable, and Beam.
2. **Bearing Load** – force distributed as a cosine pressure over the loaded half of the hole boundary, simulating a frictionless pin contact.
3. **Contact** – the rigid cylinder is explicitly modelled as a separate body; contact between cylinder and beam hole is solved nonlinearly, first without friction and then with friction ($\mu = 0.2$).

For all three methods, mesh convergence is demonstrated for displacement and von Mises stress, and failure to converge (where it occurs) is explained. All problem data required for reproducibility are given in Section 2.

2. Problem Setup and Reference Solution

2.1 Geometry and Material Properties

The beam geometry and loading are shown in Fig. 1. All dimensions and material data are listed in Table 1.

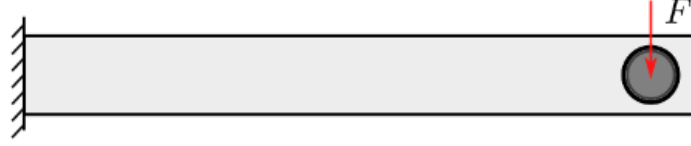


Figure 1: Cantilever beam geometry (not to scale). Fixed support on the left face; downward force $F = 2000$ N applied through the hole at $x = 185$ mm from the fixed end.

Table 1: Geometric and material parameters used throughout this report.

Parameter	Symbol	Value	Unit
Beam length	L	200	mm
Beam height	H	40	mm
Beam thickness (OOP)	t	5	mm
Hole radius	r	7	mm
Hole centre (from origin)	–	(185, 20)	mm
Applied force	F	2000	N
Young’s modulus	E	200×10^3	MPa
Poisson’s ratio	ν	0.3	–
Shear modulus ($E/2(1 + \nu)$)	G	76 923	MPa

Justification for Plane Stress

The thickness-to-height ratio $t/H = 5/40 = 0.125 \ll 1$ confirms that the structure is geometrically thin relative to its in-plane dimensions. Under this condition, the out-of-plane stress $\sigma_z \approx 0$ and the plane stress idealisation is appropriate [1].

ANSYS Model Setup

The geometry was created as a 2D surface body in DesignModeler (XY plane). In ANSYS Mechanical, **Geometry** $\rightarrow$ **2D Behavior** was set to *Plane Stress* with thickness $t = 5$ mm. A Fixed Support boundary condition was applied to the left vertical edge. The mesh used 8-noded quadrilateral elements (PLANE183) with Quadrilateral-Dominant method.

3. Method 1: Remote Force

3.1 Method Description

The Remote Force applies $F = 2000$ N (downward, $-Y$ direction) to a virtual remote point at the hole centre (185, 20, 0) mm. ANSYS internally connects this point to the scoped hole-edge nodes using constraint equations whose stiffness is controlled by the **Behavior**

setting. Four variants are investigated; their physical interpretations are summarised below and illustrated in Fig. 2.

Coupled: Rigid Multi-Point Constraint (MPC) equations enforce identical displacements for all hole-edge nodes relative to the remote point. The hole acts as if filled by a rigid plug – no relative deformation between nodes is permitted. This is the stiffest variant; stresses at the hole boundary are artificially elevated.

Rigid: Explicit rigid beam elements are created from each hole-edge node to the remote point, enforcing an identical kinematic constraint to Coupled but via an element-based formulation. Results are very similar to Coupled; this is also the stiffest option.

Deformable: The force is distributed as a weighted load to each hole-edge node. No kinematic constraints are imposed between nodes – the hole boundary deforms freely. This is the most physically realistic of the four Remote Force variants.

Beam: Massless beam elements connect each hole-edge node to the remote point, introducing a finite intermediate stiffness between Rigid and Deformable. The effective stiffness depends on the internally assigned beam cross-section.

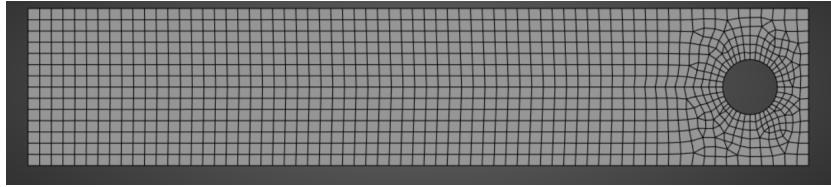


Figure 2: Cantilever beam visualisation in ANSYS. Fixed support on the left face; downward force $F = 2000$ N applied through the hole at $x = 185$ mm from the fixed end.

3.2 FE Results

Total Deformation

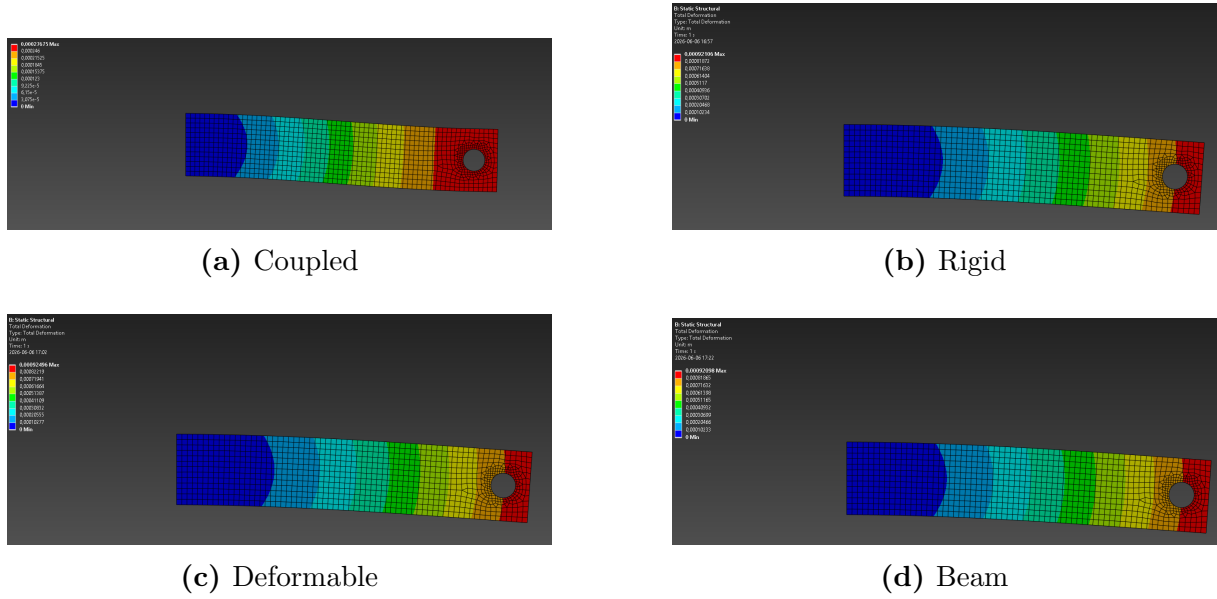


Figure 3: Total deformation contour plots for the four Remote Force behaviors (3 mm element size).

Von Mises Stress

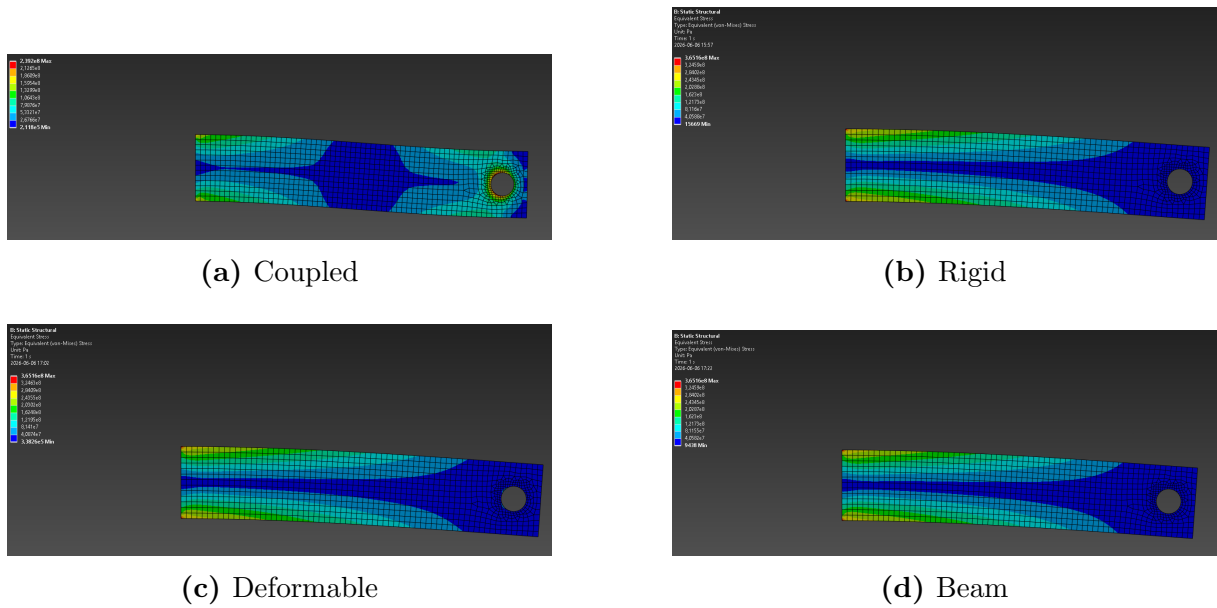


Figure 4: Von Mises equivalent stress contour plots for the four Remote Force behaviors (3 mm element size).

Numerical Comparison of Behaviors

Table 2: Comparison of Remote Force behaviors at 3 mm element size.

Behavior	Max Def. (mm)	Max σ_{VM} (MPa)
Coupled	0.276	239.2
Rigid	0.921	365.16
Deformable	0.921	365.16
Beam	0.925	365.16

Results:

As shown in Figs. 3–4 and summarised in Table 2, except coupled all three behaviors produce similar global deformations.

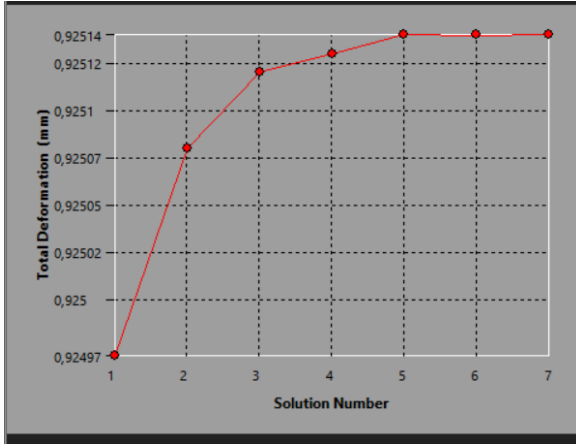
The stress distributions, however, differ markedly near the hole. For the Coupled behaviors, the MPC/rigid-link constraints impose a kinematic singularity at the hole nodes, producing unrealistically high and concentrated stresses arranged in a spoke pattern radiating from the constrained nodes. The Deformable behavior yields a smooth, physically plausible stress field around the hole, as the nodes are free to deform independently. The Beam behavior gives an intermediate distribution.

Comments:

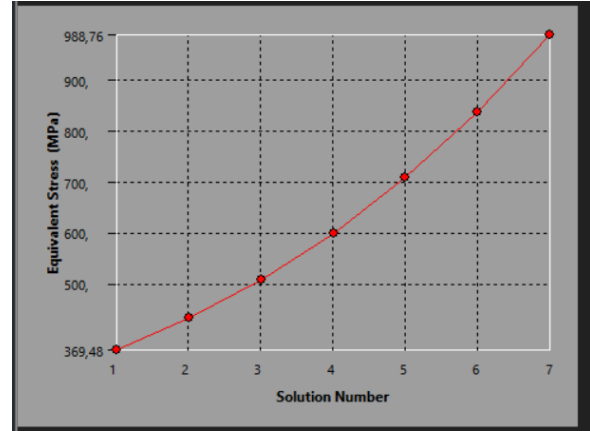
Coupled and Rigid are unsuitable when accurate local stresses near the hole are required, because the artificial constraint stiffness distorts the local stress state. Deformable is the recommended choice for general-purpose Remote Force analysis. The global displacement is insensitive to the behavior choice because the constraint region is small relative to the overall structure.

3.3 Mesh Convergence – Method 1 (Deformable)

The Deformable behavior was selected for the convergence study as the most physically representative variant.



(a) Total deformation convergence



(b) Von Mises stress convergence

Figure 5: Mesh convergence study for Method 1 (Remote Force loading). Total deformation converges smoothly, while the local peak stress near the hole boundary shows mesh sensitivity due to stress concentration.

Comments:

The total deformation converges smoothly with mesh refinement because displacement is a global response quantity controlled by the overall stiffness of the beam. As the element size decreases, the finite element approximation of the deformation field improves and approaches a stable solution.

However, the maximum von Mises stress does not show convergence when evaluated at the corners. The corners introduce a strong geometric stress concentration, and the idealized sharp edge in the finite element model creates a mesh-dependent stress increase. Therefore, finer meshes continue to capture higher local stresses near the beam edge.

4. Method 2: Bearing Load

4.1 Method Description

A Bearing Load distributes the applied force as a cosine pressure profile over the loaded half of the hole boundary. The pressure at any angular position θ (measured from the load axis) is:

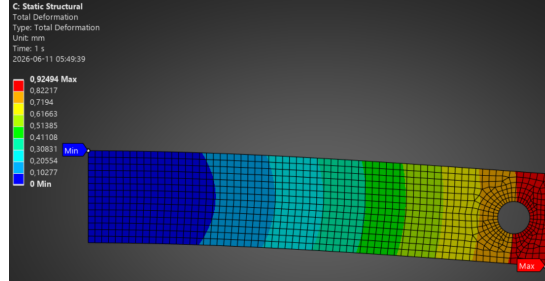
$$p(\theta) = p_0 \cos \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \quad (1)$$

The peak pressure p_0 is determined from force equilibrium:

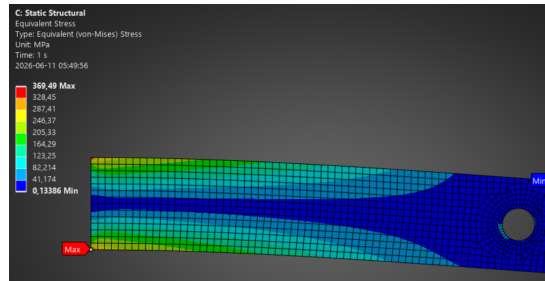
$$F = \int_{-\pi/2}^{\pi/2} p_0 \cos \theta \cdot r t d\theta = 2 p_0 r t \implies p_0 = \frac{F}{2rt} = \frac{2000}{2 \times 7 \times 5} = 28.57 \text{ MPa} \quad (2)$$

The load is applied to the lower semicircle, consistent with the downward force direction. In ANSYS, the full hole boundary is scoped and the software automatically determines the loaded half from the specified force direction. This method simulates a frictionless pin-in-hole interaction without explicitly modelling the pin geometry.

4.2 FE Results



(a) Total deformation



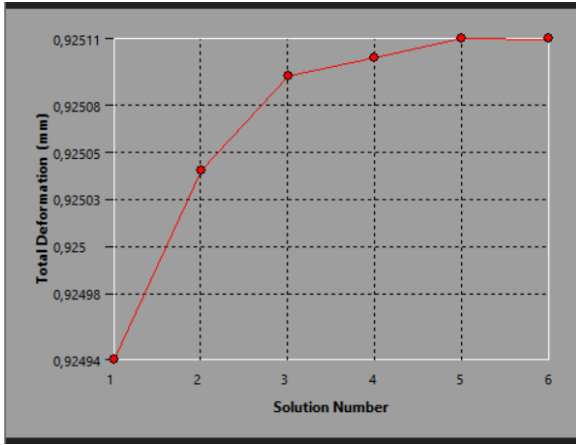
(b) Von Mises stress

Figure 6: Bearing Load results (3 mm element size).

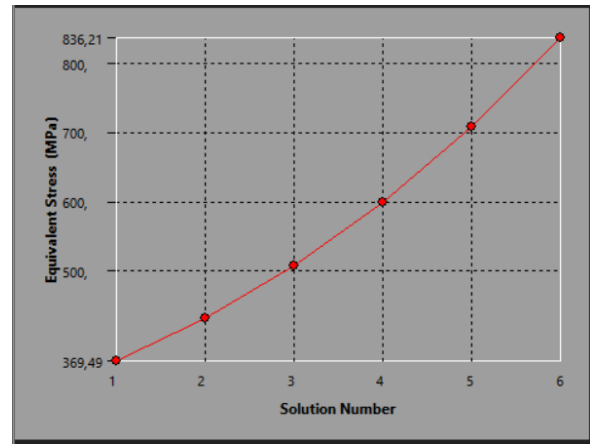
Results:

Fig. 6 confirms that the cosine pressure is correctly applied to the lower semicircle with maximum intensity at the bottom of the hole ($\theta = 270$), tapering to zero at $\theta = \pm 90$, and zero on the upper half. The global deformation and the far-field stress distribution are very close to those from the Remote Force (Deformable) result - the two methods are approximating the same physical load transfer mechanism.

4.3 Mesh Convergence – Method 2



(a) Total deformation convergence



(b) Von Mises stress convergence

Figure 7: Mesh convergence study for Method 2 (Bearing Load). The displacement converges with mesh refinement, whereas the peak stress near the bearing load boundary remains mesh dependent.

The total deformation converges consistently with mesh refinement because the bearing load provides a distributed load over the hole surface, resulting in a stable global deformation response. Once the mesh is sufficiently refined, further mesh refinement produces negligible changes in the overall displacement.

The maximum von Mises stress near the hole boundary does not converge due to the presence of a local stress concentration at the bearing load application region. The transition between the loaded and unloaded portions of the hole creates a highly localized stress field, which becomes increasingly captured as the mesh is refined. This causes the peak stress value to continue increasing. Therefore, the stress convergence evaluation is performed using a far-field stress probe away from the hole where the stress distribution is smooth.

5. Method 3: Contact

5.1 Method Description

Method 3 explicitly models the rigid cylinder as a second surface body inside the beam hole and solves the contact problem nonlinearly. Two sub-cases are studied:

1. **Frictionless:** only compressive normal forces are transmitted ($\tau = 0$).
2. **Frictional** ($\mu = 0.2$): tangential Coulomb friction forces are also transmitted; sliding occurs when $|\tau| = \mu |\sigma_n|$.

This is the most physically complete method – the contact pressure distribution and contact area emerge from the geometry and material without any assumed distribution. It is also the most computationally expensive due to the nonlinear iteration required.

Additional Geometry in DesignModeler

A second surface body (circle, $r = 7$ mm, centre (185, 20) mm) was added in DesignModeler as **Sketch2** on the XY plane, with dimensions exactly matching the beam hole. The feature *Surfaces From Sketches* was applied to Sketch2 alone, resulting in two bodies: the beam and the cylinder disc.

Contact and Boundary Condition Settings

In ANSYS Mechanical the following settings were used (Table 3). The cylinder was assigned **Stiffness Behavior = Rigid**; its pilot node is located at (185, 20, 0) mm.

Table 3: Contact region settings for Method 3.

Parameter	Frictionless	Frictional
Type	Frictionless	Frictional
Friction coefficient	–	0.20
Contact body	Beam hole inner edge (flexible)	
Target body	Cylinder outer edge (rigid)	
Behavior	Asymmetric	
Formulation	Augmented Lagrange	
Detection method	Nodal-Normal to Target	
Interface treatment	Adjust to Touch	
<i>Loads on rigid cylinder (Remote, Behavior = Rigid):</i>		
Remote Force	$F_y = -2000$ N at (185, 20, 0) mm	
Remote Displacement	$X = 0$, $Y = \text{Free}$, $\text{Rot}_Z = 0$	

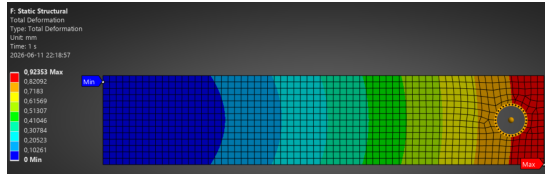
5.2 FE Results – Frictionless Contact



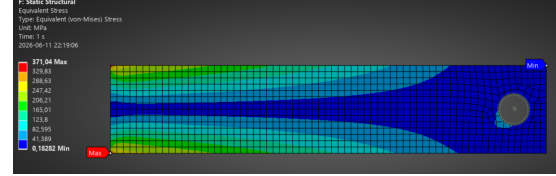
(a) Contact status (Closed / Open)



(b) Contact pressure



(c) Total deformation



(d) Von Mises stress

Figure 8: Method 3 – frictionless contact results (3 mm element size).

Results:

As shown in Fig. 8, the contact status plot confirms that closed contact (compressive normal force) occurs on the lower arc of the hole, while the upper arc is open, in full agreement with the expected physics of a downward-loaded cylinder. The contact pressure profile follows an approximately cosine shape, qualitatively matching the prescribed distribution of Method 2 (Bearing Load).

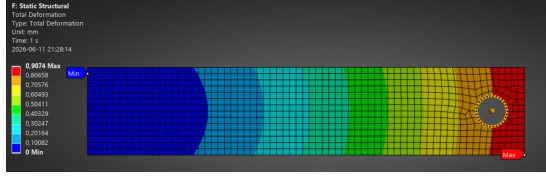
5.3 FE Results – Frictional Contact ($\mu = 0.2$)



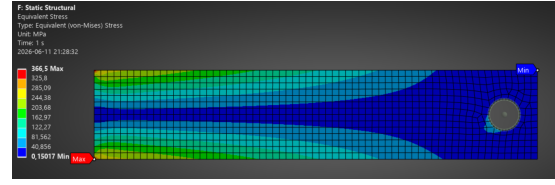
(a) Contact pressure



(b) Frictional Contact Status



(c) Total deformation



(d) Von Mises stress

Figure 9: Method 3 – frictional contact ($\mu = 0.2$) results (3 mm element size).

Results:

Introducing friction ($\mu = 0.2$) produces non-zero tangential stresses at the contact interface (Fig. 9b). The contact pressure distribution shifts slightly relative to the frictionless case, altering the equilibrium state, confirming that friction at this coefficient has a limited effect on the structural response but is significant for the local stress state at the contact zone.

5.4 Mesh Convergence – Method 3

Contact Sizing was applied to the hole and cylinder edges to maintain adequate mesh density at the contact interface.

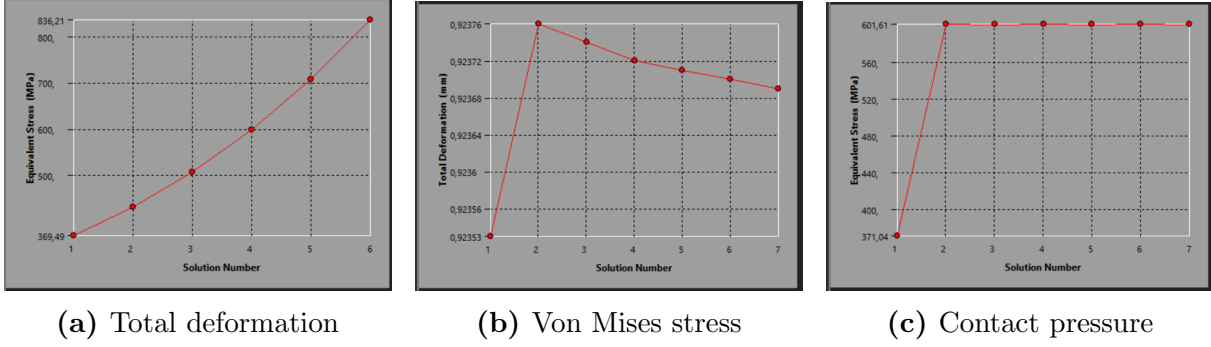


Figure 10: Mesh convergence study for Method 3 (Frictionless contact). Deformation and far-field stress converge, while the maximum contact pressure remains mesh dependent due to the contact-edge singularity.

Comments:

Both the total deformation and the far-field von Mises stress converge smoothly with mesh refinement because the contact formulation correctly transfers the applied load from the rigid cylinder to the deformable beam. The global structural response becomes independent of mesh density once sufficient refinement is achieved.

However, the maximum contact pressure does not converge because a contact-edge singularity exists at the transition between the closed-contact and open-contact regions. At this location, the boundary condition changes abruptly from a contact constraint to a free surface, producing an idealized unbounded pressure concentration in the continuum solution. As the mesh is refined, ANSYS captures higher local pressure values at this transition region. Therefore, the peak contact pressure is considered mesh-dependent and is not used as the primary convergence criterion. The convergence assessment is instead based on total deformation and far-field stress away from the contact boundary.

6. Overall Comparison and Discussion

6.1 Numerical Comparison

Table 4 presents a consolidated comparison of all methods at a uniform element size of 1.5 mm. All probe locations are identical across runs to enable a direct comparison.

Table 4: Consolidated numerical comparison of all methods (3 mm element size).

Method	Max Def. (mm)	Max σ_{VM} (MPa)
Remote Force – Coupled	0.276	239.2
Remote Force – Rigid	0.921	365.16
Remote Force – Deformable	0.921	365.16
Remote Force – Beam	0.925	365.16
Bearing Load	0.925	369.49
Contact – Frictionless	0.923	371.04
Contact – Frictional ($\mu = 0.2$)	0.907	366.5

6.2 Applicability and Relative Merits

Remote Force (Deformable): The most convenient method for global structural response when local stresses near the hole are not the primary interest. No additional geometry is required and setup is fast. Deformable is the only acceptable Behavior choice for stress analysis; Coupled and Rigid introduce artificial stiffness and unphysical stress concentrations at the constrained nodes. Remote Force cannot capture friction or represent the true contact pressure distribution.

Bearing Load: A physically motivated and efficient method for pin-in-hole loading. The cosine distribution is the classical frictionless contact solution and requires no pin geometry. It gives more realistic local stresses than any Remote Force variant. Limitations: friction cannot be modelled, partial contact (e.g. due to bending or offset loading) is not captured, and the stress singularity at the loaded/unloaded transition prevents stress convergence at those nodes.

Contact (Frictionless and Frictional): The most physically complete method. The contact area, contact pressure profile, and frictional stress all emerge directly from the geometry and loading without any assumed distribution. Friction can be included. The method is also the most expensive: it requires additional geometry (the cylinder body), a nonlinear solver, and more complex setup. It is the method of choice when local stress accuracy near the hole is critical or when the assembly includes friction. A contact-edge singularity prevents stress convergence at the contact boundary transition, but this is a fundamental continuum mechanics result and not a modelling error.

References

- [1] S. P. Timoshenko and J. N. Goodier. *Theory of Elasticity*. McGraw-Hill, 1970.